Reflection on Image Processing Tasks

Working through these tasks made me appreciate how different representations and algorithms reveal distinct aspects of an image, and why choosing the right tool matters.

Colour Spaces

Exploring HSV and LAB channels highlighted how each encodes colour differently:

- **HSV** separates colour identity (Hue) from intensity (Saturation) and brightness (Value). This makes it far more robust for segmentation under varying lighting conditions, since the Hue remains stable even when shadows or highlights change.
- **LAB** decouples lightness from chromatic components (a: green to red, b: blue to yellow), which is useful for perceptual uniformity and colour correction.

This reinforced why HSV is often preferred for colour-based object detection: it isolates the "what colour is it?" question from "how bright is it?" or "how vivid is it?"

Noise and Filtering

The PSNR experiments showed how filter choice depends on noise type:

- **Gaussian noise** was best handled by a Gaussian filter, since its kernel matches the statistical distribution of the noise, smoothing fluctuations while preserving structure.
- **Salt-and-pepper noise** was best handled by a median filter, which discards extreme outliers naturally. Box and Gaussian filters struggled because they averaged in the corrupted pixels.

This demonstrated the importance of matching the filter to the noise characteristics rather than assuming one universal solution.

Edge Detection

Testing thresholds in Canny revealed how hysteresis works:

- Low thresholds keep many weak edges (including noise).
- Balanced thresholds preserve strong edges while suppressing isolated weak ones.
- High thresholds yield a sparse but clean edge map.

Comparing Sobel and Laplacian of Gaussian (LoG) further clarified their roles: Sobel, as a first-derivative operator, emphasizes direction and strength of intensity changes, while LoG, as a second-derivative operator, produces thinner, symmetric outlines.

Morphology and Object Counting

The morphology pipeline showed how structuring element size and shape directly affect results. Using a 3×3 elliptical kernel balanced noise removal with object preservation, preventing coins from merging. The minimum area filter of 70 pixels ensured only blobs large enough to be coins were counted, eliminating hundreds of tiny fragments.

This highlighted the delicate balance between cleaning masks and preserving distinct objects — too aggressive a kernel merges coins, too small an area threshold counts noise.

Failure Mode

The morphology pipeline failed to count coins using the same elliptical kernel and minimum area if the new image introduced had different sized coins.